

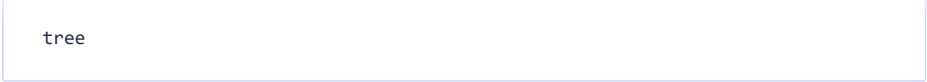
```
main.py x
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  @app.route('/')
6  def birthday():
7      return 'Hello from Flask'
8
9  if __name__ == '__main__':
10     app.run(host='0.0.0.0', port=3000, debug=True)
```

Guide x

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Flask App



The `birthday` directory represents the Flask app. The Python logic is in `main.py` (often `app.py` is used instead), the `static` directory contains our CSS to style the site, and the `templates` directory contains all of our HTML templates. You *must* have a directory called templates as Flask expects this to exist.



Starting Our Flask App

We will begin with our `main.py` Python file. Click the link below to open it.

Open main.py

Copy and paste the starter code into the file.

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def birthday():
    return 'Hello from Flask'

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=3000, debug=True)
```

We are importing `Flask` from its page and then instantiating a Flask object. The `app.route` decorator is used to define the different views for our app. The `/` view is the first content you will see when the page loads. For now, return a string with a greeting. Finally, when we call this script, run our Flask app. **Important**, host and port have specific information so they work on Codio. If you are developing a Flask app on your own machine, Flask will provide defaults that will work just fine.

Return to the terminal by clicking on the appropriate tab in the left panel. Then run the Flask app. Remember, the `main.py` is in the `birthday` directory, so we need to specify this with our Python command (or we need to change into the correct directory).

```
python birthday/main.py
```

You will see terminal output about a development server running. It is running port 3000 of our local host, specifically the address `0.0.0.0`. Click the link below to preview our website. You should see our simple greeting and nothing more.

Open Flask app

► Did you notice?

Try this variation:

- Change the greeting in our Flask app by following these steps:
 - Stop the Flask app by going to the terminal and pressing the `Ctrl` and `C` keys at the same time. If you are on a Mac it would be `control` and `C`.
 - Click on the tab for the `main.py` file.
 - Change the message in the `return` statement to be whatever you like.
 - Go back to the terminal and run the Flask app again with:

```
python main.py
```

- Click on the tab with the Flask preview and click on the icon with the blue arrows to refresh the preview.
- You should see your new message.

In the terminal, stop the Flask app with `ctrl + c` and then deactivate the `birthday` virtual environment.

```
conda deactivate
```

Reading Question

Fill in the blanks below.

- Flask apps use `HTML, CSS, and JavaScript` on the front end.
- Flask apps use `Python` on the back end.
- Flask apps allow for the reuse of HTML code with `templates`.

The correct answers are:

- Flask apps use **HTML, CSS, and JavaScript** on the front end.
- Flask apps use **Python** on the back end.
- Flask apps allow for the reuse of HTML code with **templates**.

For the front end, Flask relies on HTML, CSS, and JavaScript to build and style webpages. On the back end, however, Python is used for the logic (setting routes) and manipulation of data. Flask speeds up development time with its templating system. You can reuse HTML documents but plug in new information. This keeps you from having to write several documents that are almost identical.

Check It!

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